

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

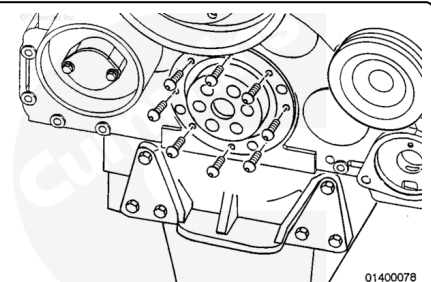
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Remove the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/20/20-008-040.html>).
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>).
- Remove the vibration damper and fan drive pulley. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html>).

Remove

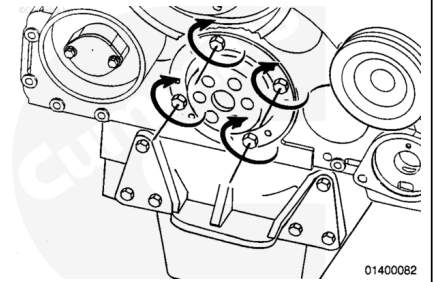
Remove the eight button head hex socket capscrews.



Use four [1/4-20 UNCx1 in] capscrews to remove the seal from the crankshaft.

Install the capscrews into the clinch nuts in the seal case.

Tighten the capscrews alternately and evenly to remove the seal.

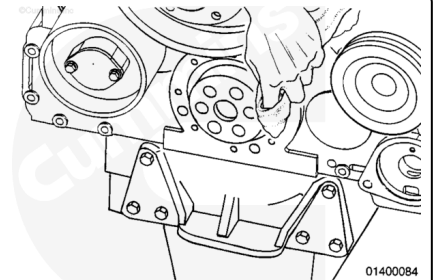


Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Use a clean cloth and safety solvent, Part Number 3823717, or equivalent.

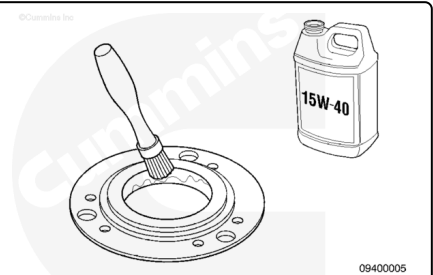


Install

To aid in seal installation, use a small nonmetallic bristle brush to apply a thin film of SAE 15W-40 oil to the inside diameter of the seal case.

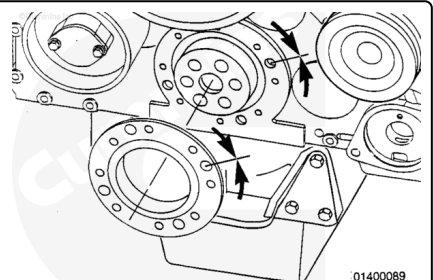
⚠ CAUTION ⚠

Do not allow oil to come in contact with other areas of the seal to avoid damage to sealing surfaces.

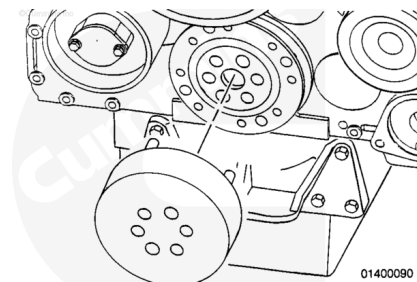


Use hand pressure to push the seal on the crankshaft as far as possible.

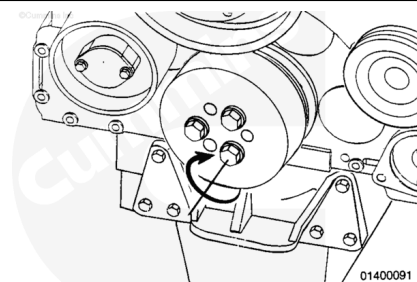
Align the four clinch nuts on the back of the seal carrier flange with the counterbores in the gear cover mounting surface.



Install the seal installation tool that is supplied with the seal on the crankshaft. Position the installation tool locating pins in the seal mounting capscrew holes.



Use three crankshaft adapter mounting capscrews. Tighten the capscrews alternately and evenly in one-half of a turn increments until the seal flange contacts the gear cover.

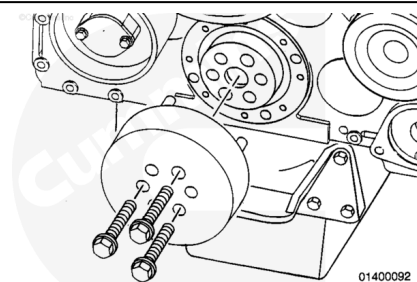


⚠ CAUTION ⚠

Do not continue to tighten the capscrews once the seal flange contacts the gear cover. Over-tightening will result in gear cover damage.

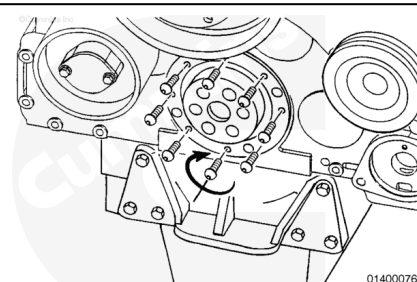
Remove the installation tool.

Note : It is normal for the seal flange to spring away from the front cover when the installation tool is removed. Do not attempt to seal the flange again as gear cover and seal damage may result.



Install the eight button head hex socket capscrews. Tighten the capscrews alternately and evenly.

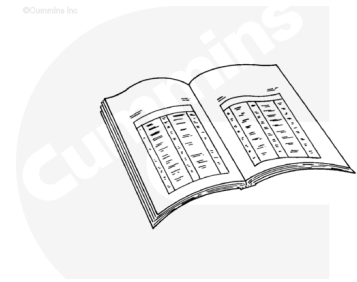
Torque Value: 11 n·m [97 in-lb]



Finishing Steps

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- Install the vibration damper and fan drive pulley. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/20/20-001-052-tr.html>).
- Install the fan spacer(s) and fan. Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/20/20-008-040.html>).
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>).

Operate the engine and check for leaks.